

Plant Biotechnology

Lab. BT 421

Lab (7)

Protoplast Isolation

By:

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Outline

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- ❑ **Protoplast applications**
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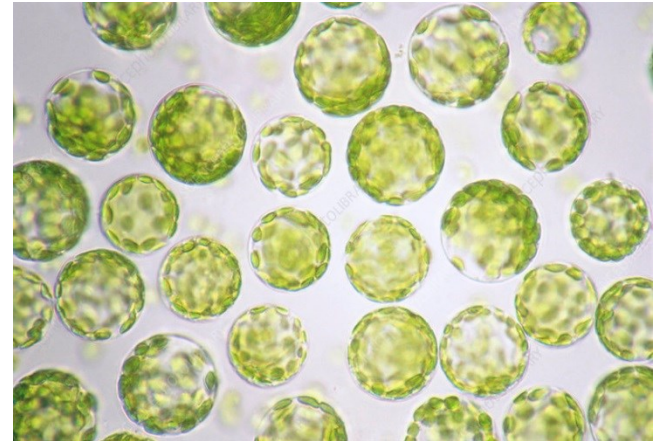
Protoplast

- ❑ **Plant protoplast:** spherical naked plant cells obtained by removal or digestion of cell wall (Partially or completely).
- ❑ Protoplast term was introduced in 1880 by **Hanstein**.



Polyhedral shape

Plant cells
→
After cell wall
removal



Spherical shape

Fragile and fully exposed plasma
membrane

The Cell wall

- ❑ **Cell wall:** thick layer found outside the cell membrane and surrounds the cell.
- ❑ Cell wall provides structural support and protection by preventing the plant cell from bursting when water enters the cell.



The Cell wall

The Cell wall

- ❑ The plant cell wall is made of **cellulose fibers** forming with **hemicelluloses** a network which is embedded in the pectin matrix.
- ❑ The adjacent cells are linked by the middle lamella mainly composed of **pectin** (Figure 1).

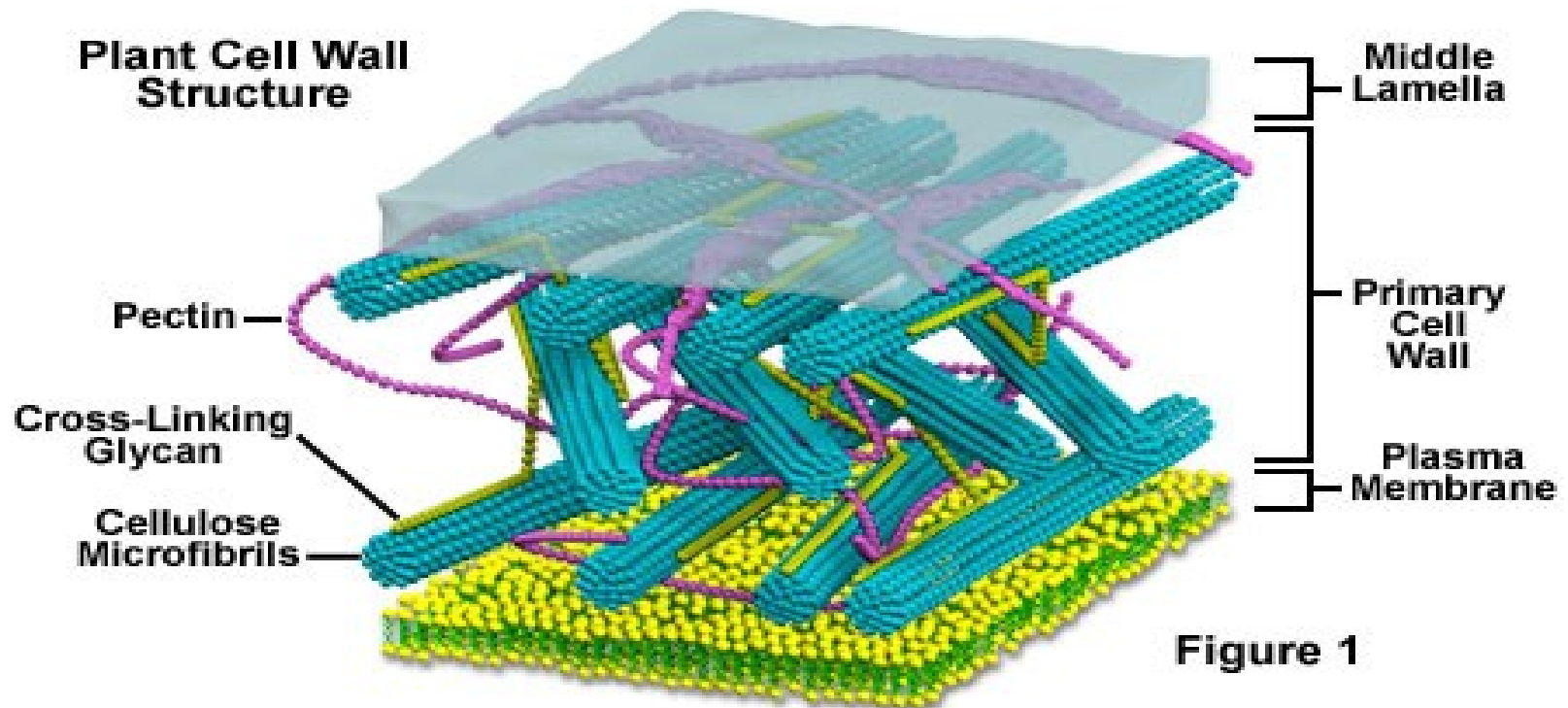


Figure 1

Protoplast applications

There are many applications for protoplast, but here are some of them:

- ☐ **Crop improvement by somatic hybridization to produce new crop that is a stress tolerant.**
- ☐ **Introducing new hybrids from different species fusion.
(Potato + Tomato = Pomato)**
- ☐ **Producing new products for industrial uses (Rubber).**

Protoplast application

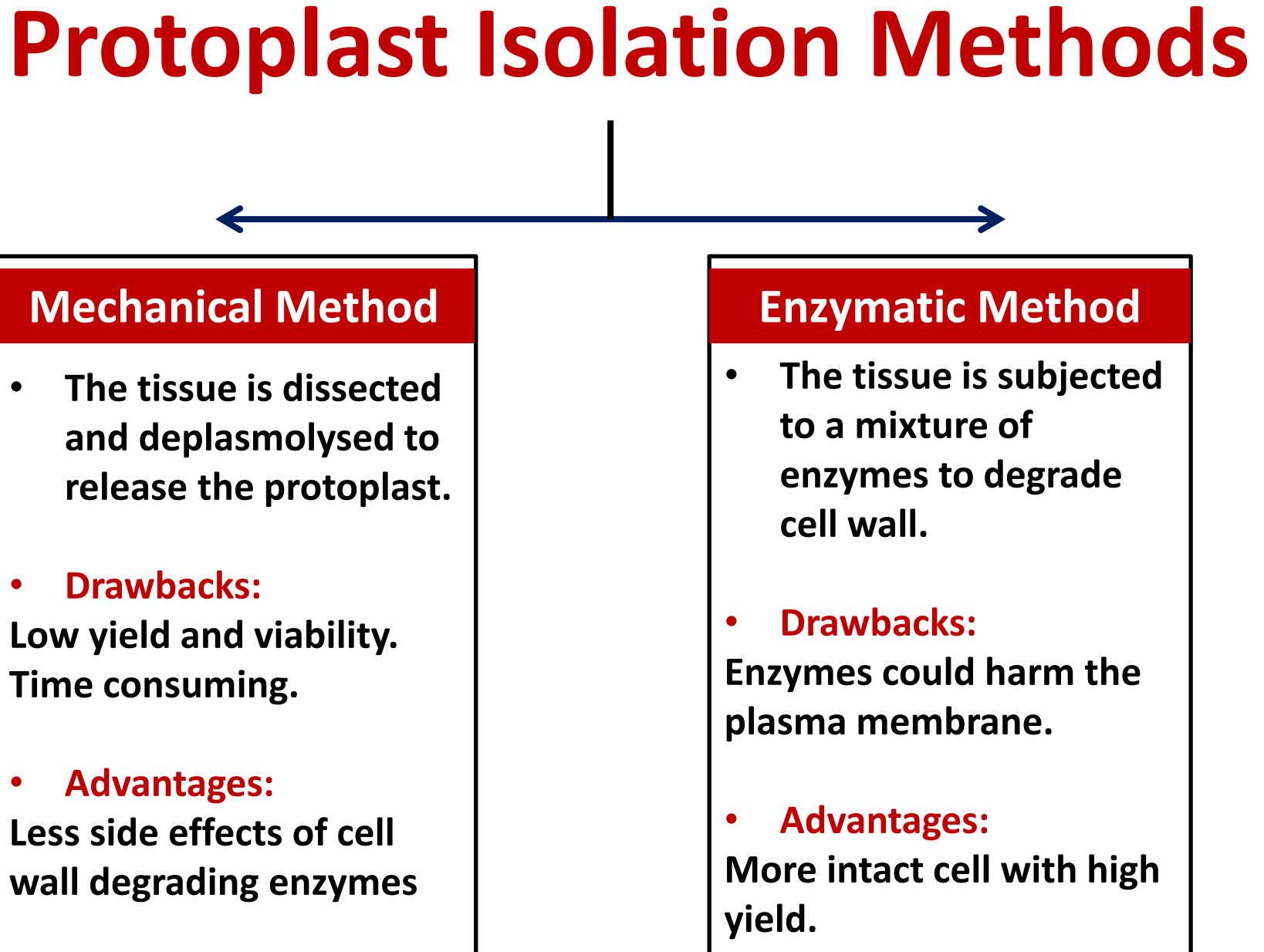
❑ **Implantation of Chloroplast.**

❑ **Transplantation of Nuclei and chromosomes.**

Isolated nuclei can be introduced into the protoplasts. It is really opening up new avenues for the study of nuclear- cytoplasmic interaction if fertile plants with foreign nuclei could be regenerated from such protoplasts.

This procedure provides a valuable method for genetic information transfer and gene analysis.

Protoplast Isolation Methods



Mechanical Method

- The tissue is dissected and deplasmolysed to release the protoplast.
- **Drawbacks:**
Low yield and viability.
Time consuming.
- **Advantages:**
Less side effects of cell wall degrading enzymes

Enzymatic Method

- The tissue is subjected to a mixture of enzymes to degrade cell wall.
- **Drawbacks:**
Enzymes could harm the plasma membrane.
- **Advantages:**
More intact cell with high yield.

How to isolate protoplast

Mechanically

The isolation of protoplast from mesophyll cells consists of the following steps:

1. Surface sterilization.

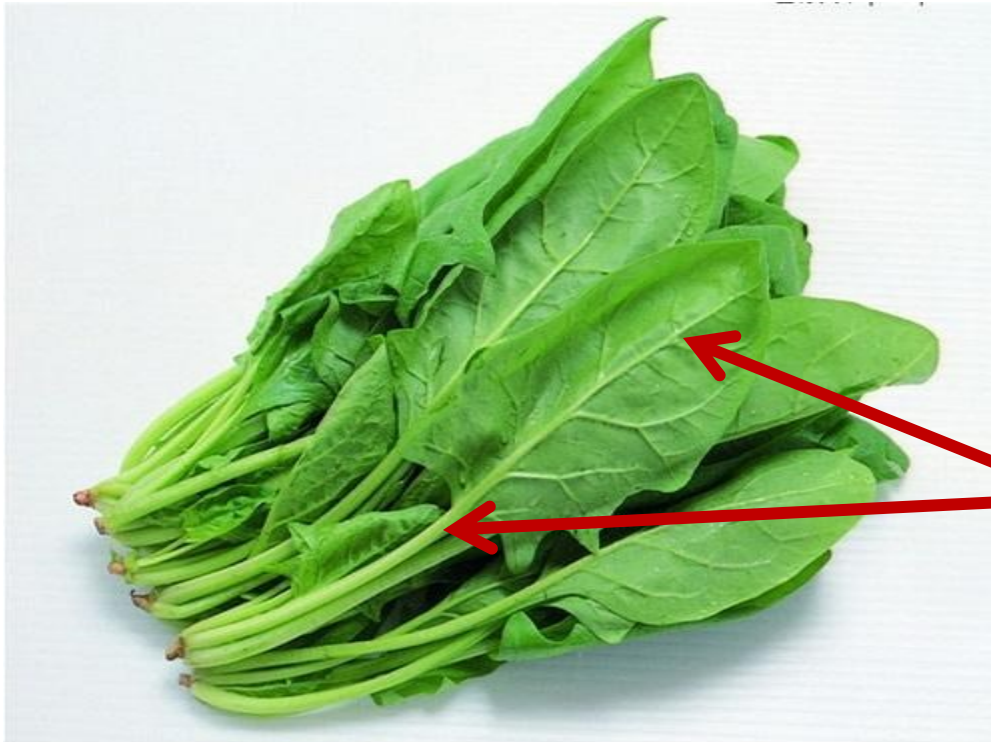
The leaves obtained from 7-8 week old plants are surface sterilized (we will use spinach).

Sterilization is done by immersing the leaves in 70% ethyl alcohol for 30-60 seconds followed by rinsing in 0.4-0.5% sodium hypochlorite solution (Hypex) for 15-20 minutes.

How to isolate protoplast

Mechanically

2. Wash the leaves repeatedly with sterile distilled water and remove the veins.

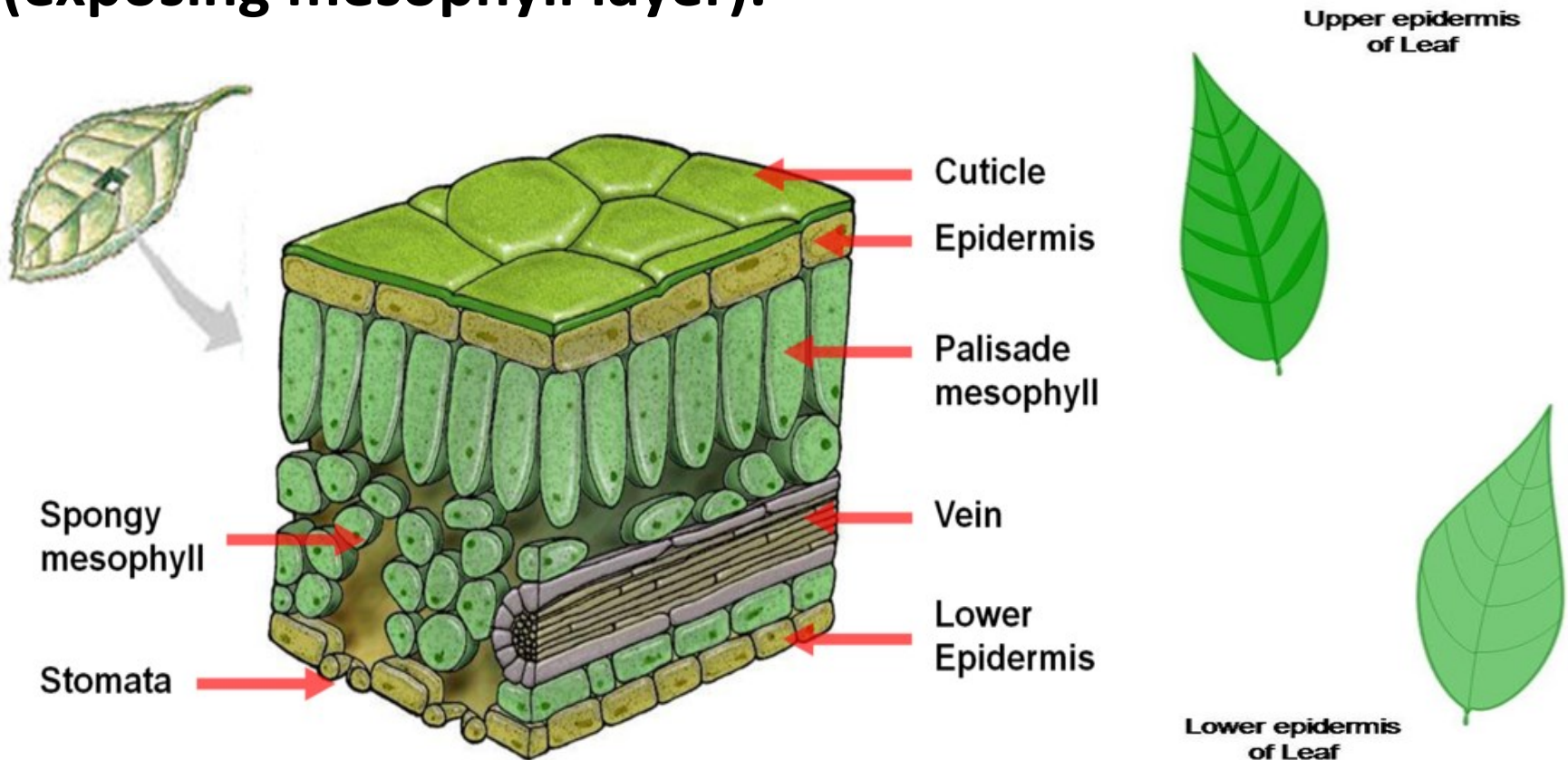


(The veins)
Vascular bundles

How to isolate protoplast

Mechanically

3. Peel the lower epidermis carefully using a forceps (exposing mesophyll layer).



How to isolate protoplast

Mechanically

4. Fragmentation:

Cut the plant tissue into smaller pieces using a sharp knife (increasing the surface area).

5. Grind the leaves with **CPW buffer** (4-5 ml) gently using pestle and mortar (Destroying the cell wall).



How to isolate protoplast

Mechanically

6. Soak the leaves with (CPW solution + 10% mannitol) or sorbitol for 20 minutes at room temperature.

(10 grams of mannitol in 100 ml of CPW solution)

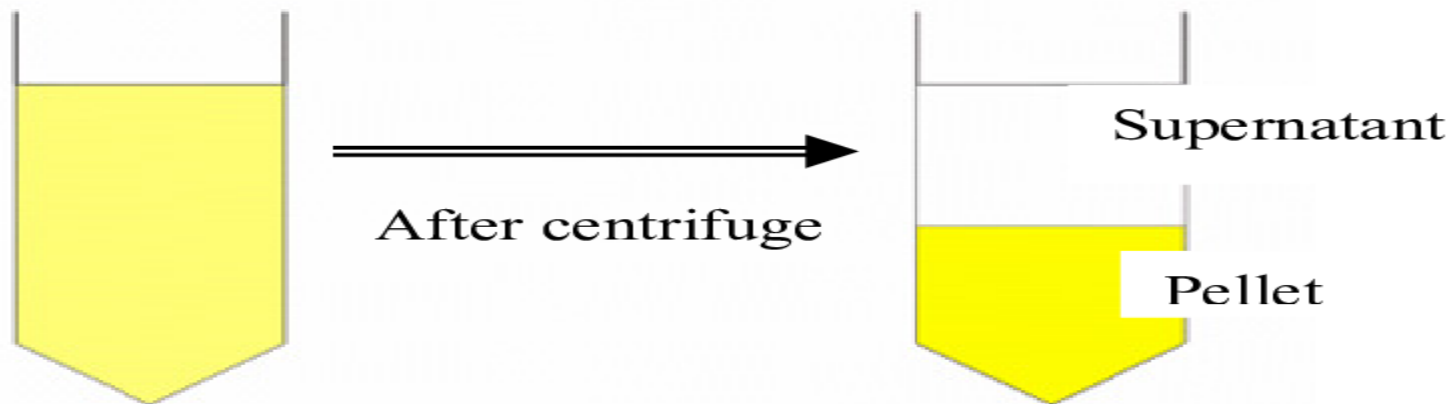
- **Mannitol** maintains a slightly hyperosmotic environment protecting protoplast from burst.

How to isolate protoplast

Mechanically

7. Filter the leaves and take the extract by using cheese cloth.

8. Centrifuge the extract (3000 rpm/ 10 min.) and then discard the supernatant.

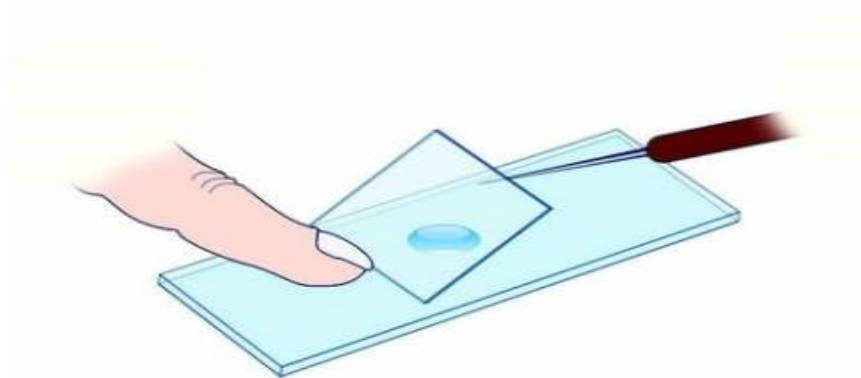


How to isolate protoplast

Mechanically

9. Suspend the pellet in CPW and mannitol by adding this solution to the centrifuge tube.

10. Prepare a wet mount slide and examine under the microscope (1000X).



How to isolate protoplast

Enzymatically

❑ The first four steps in Enzymatic method are the same in the mechanical one.

5. Place the peeled leaf pieces in enzyme mixture containing cellulase, hemicellulase, and pectinase and shaken in a water bath at 25 °C.

How to isolate protoplast

Enzymatically

6. Replace the enzyme mixture with another fresh one in addition to mannitol at pH 5.4 and incubate for 12 hours at 25 °C.

7. Filter the protoplast gently to remove the debris.

8. Centrifuge the filtrate and discard the supernatant.

How to isolate protoplast

Enzymatically

9. Wash the pellet with 0.7 M sorbitol and later replaced with 20% sucrose solution.

(You can keep the protoplasts in isotonic solution)

10. Prepare a wet mount slide.

11. Examine under the microscope.

Protoplast under microscope

- ❑ Protoplasts in culture generally start to **regenerate** a cell wall within few hours after isolation and may take two to several days to complete the process under suitable conditions.

